

## CYBR 343 Test3

### Instructions:

- Type or write your answers directly below each question.
- Show all work and clearly explain your reasoning.
- Each response must be 4–8 lines in complete sentences.
- Include a screenshot for every question showing supporting evidence.
- Place each image directly under the response.
- Ensure all images are clear and properly oriented.
- Submit via Blackboard only. Email submissions are not accepted.
- I am available for questions until Friday 2:00 PM only.
- No credit without both explanation and screenshot evidence.

### Q1. 30 Points

Using Registry Ripper and the provided SAM and SYSTEM registry hives obtained from a computer, answer the following:

1. Identify the user who has completed their security questions.
  - What is the username?
    - The username is BlueStar [User 1004]
  - What security questions are configured?
    - The following questions are configured as security questions:
      - What was your first pet's name?
      - What's the name of the city where you were born?
      - What was your childhood nickname?
  - What are the corresponding answers?
    - Q1: BlueCat
    - Q2: BlueCity
    - Q3: Blue

```
Username      : BlueStar [1004]
Full Name     :
User Comment  :
Account Type  :
Account Created : 2026-04-05 01:24:11Z
Name          :
Last Login Date : Never
Pwd Reset Date : 2026-04-05 01:24:11Z
Pwd Fail Date  : Never
Login Count   : 0
Embedded RID   : 1004
--> Password does not expire
--> Password not required
--> Normal user account

ForcePasswordReset : 0
Security Questions:
  Question 1 : What was your first pet's name?
  Answer      : BlueCat
  Question 2 : What's the name of the city where you were born?
  Answer      : BlueCity
  Question 3 : What was your childhood nickname?
  Answer      : Blue
```

2. Determine the network configuration for the system.

- What IP address is assigned to the Ethernet interface?
  - 10.0.2.15
- What is the DHCP server address?
  - 10.0.2.2
- What are the lease obtained and lease expiration times?
  - Lease Obtained Time: 2026-04-05 @ 00:57:41Z (UTC)
  - Lease Expiration (Terminate) Time: 2026-04-06 @ 00:57:41Z
- Include any additional relevant network details.
  - Subnet Mask: 255.255.255.0
  - Lease duration: 24 Hours
  - DHCP is enabled

```
nic2 v.20200922
(System) Gets NIC info from System hive

Adapter: {0e9c9087-e30c-4d32-8acd-e9f004fe1fcd}
LastWrite Time: 2026-04-05 00:57:41Z
EnableDHCP 1
Domain
NameServer
DhcpIPAddress 10.0.2.15
DhcpSubnetMask 255.255.255.0
DhcpServer 10.0.2.2
Lease 86400
LeaseObtainedTime 2026-04-05 00:57:41Z
T1 2026-04-05 12:57:41Z
T2 2026-04-05 21:57:41Z
LeaseTerminatesTime 2026-04-06 00:57:41Z
AddressType 0
IsServerNapAware 0
DhcpConnForceBroadcastFlag 0
DhcpInterfaceOptions
005 0 000 0 3 0 000 0000
000 000 home.local 0 0 000 00 00 000
000 0 000 000
DhcpDomain home.local
DhcpNameServer 192.168.7.1
DhcpDefaultGateway 10.0.2.2
DhcpSubnetMaskOpt 255.255.255.0
DhcpGatewayHardware
000 RT 050
DhcpGatewayHardwareCount 1

ControlSet001\Services\Tcpip\Parameters\Interfaces has no subkeys.
Adapter: {1d7be783-d1fc-11f0-a1b4-806e6f6e6963}
LastWrite Time: 2025-12-05 17:03:35Z

ControlSet001\Services\Tcpip\Parameters\Interfaces has no subkeys.
Adapter: {9f000b74-abd1-4474-8385-d68377cddb5a}
LastWrite Time: 2025-12-05 17:03:56Z
EnableDHCP 1
Domain
NameServer

ControlSet001\Services\Tcpip\Parameters\Interfaces has no subkeys.
```

3. Identify evidence of program execution.

- When was ipconfig.exe executed?
  - ipconfig.exe was last executed on 2025-09-15 @19:41:16

```
C:\Windows\System32\ipconfig.exe 2025-09-15 19:41:16
```

- When was Diskview64.exe executed?
  - Diskview64.exe was last executed 2025-12-06 @00:35:07

```
C:\Tools\DiskView64.exe 2025-12-06 00:35:07
```

4. Determine the last system shutdown time.

- The last system shutdown was on 2026-04-05 @00:51:11 UTC

```
shutdown v.20201005
(System) Gets ShutdownTime value from System hive

ControlSet001\Control\Windows key, ShutdownTime value
LastWrite time: 2026-04-05 00:51:11Z
ShutdownTime : 2026-04-05 00:51:11Z
```

Q2. 30 Points

Using Registry Ripper and the provided NTUSER.dat hive, answer the following:

a. Identify the user associated with this NTUSER.dat file.

- The user associated in the NTUSER.dat file is admin123

```
2025-09-15 19:41:02Z - C:\Users\admin123\AppData\Local\Microsoft\OneDrive\25.087.0506.0001_1\FileSyncConfig.exe
2025-09-15 19:41:02Z - C:\Users\admin123\AppData\Local\Microsoft\OneDrive\25.216.1104.0002\FileSyncConfig.exe
2025-09-15 19:41:02Z - C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2501.31.0_x64__8wekyb3d8bbwe\Notepad\Notepad.exe
2025-09-15 19:41:02Z - SIGN.MEDIA=16F4A600 vboxadditions\VBBoxWindowsAdditions.exe
2025-09-15 19:41:02Z - C:\Users\admin123\Downloads\ChromeSetup.exe
2025-09-15 19:41:02Z - C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_25.2.240.0_x64__3847v3x7pw1km\py-manager.exe
2025-09-15 19:41:02Z - C:\Tools\Exterro_FTK_Imager_(x64).exe
2025-09-15 19:41:02Z - C:\Tools\ImagerPro_8.2.0.26\Exterro_FTK_Imager_(x64).exe
2025-09-15 19:41:02Z - C:\Program Files\CodeMeter\Runtime\bin\CodeMeterCC.exe
2025-09-15 19:41:02Z - C:\Windows\Installer\MSIF59.tmp
2025-09-15 19:41:02Z - C:\Windows\System32\msiexec.exe
2025-09-15 19:41:02Z - C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_25.2.240.0_x64__3847v3x7pw1km\python.exe
2025-09-15 19:41:02Z - C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_25.2.240.0_x64__3847v3x7pw1km\pymanager.exe
2025-09-15 19:41:02Z - C:\Users\admin123\Downloads\python-3.14.2-amd64.exe
```

b. List the different commands executed by this user.

- The following commands were executed by this user:
  1. Notepad – 2026-04-05 @22:15:36
  2. RegRipper – 2026-04-05 @22:14:16
  3. Windows Explorer – 2026-04-05 @22:06:39
  4. FTK Imager – 2026-04-05 @22:06:29
  5. RegRipper – 2026-04-05 @01:07:19
  6. Chrome – 2026-04-05 @01:05:00
  7. Windows Store – 2026-04-05 @00:44:49
  8. Microsoft Edge – 2026-04-05 @00:41:46
  9. FTK Imager (x64) – 2026-04-05 @00:39:58
  10. Xbox Gaming Overlay – 2026-04-04 @22:21:00
  11. VoiceAccess – 2026-03-14 @14:42:40
  12. Outlook – 2026-03-14 @14:30:12

13. Outlook (App) – 2026-03-14 @14:30:11
14. Paint – 2026-03-14 @14:30:01
15. MSPaint.exe – 2026-03-14 @14:30:01
16. Command Prompt – 2026-03-14 @00:56:08
17. UserAccountControlSettings – 2026-03-14 @00:54:52
18. Hex Workshop – 2026-01-19 @16:29:25
19. Python Installer – 2026-01-14 @00:43:15
20. Python Installer (Cache) – 2026-01-14 @00:34:06
21. msixexec – 2026-01-14 @00:33:53
22. AutoGenerated App – 2026-01-03 @02:01:22
23. SystemPropertiesAdvanced – 2026-01-03 @00:28:09
24. PowerShell – 2025-12-07 @13:59:32
25. python.exe – 2025-12-07 @13:48:20
26. Volatility – 2025-12-07 @02:46:24
27. Python Manager – 2025-12-07 @02:26:28
28. Python Manager (App) – 2025-12-06 @23:48:52
29. FTK Imager (ImagerPro) – 2025-12-06 @00:38:35
30. VBoxWindowsAdditions – 2025-12-06 @00:16:03 Feedback Hub – 2025-12-05 @14:36:56
31. Sticky Notes – 2025-12-05 @14:36:56
32. Calculator – 2025-12-05 @14:36:56

```

2026-04-05 22:15:36Z
Microsoft.WindowsHotepad_Bueky3d8bbweApp (20)
2026-04-05 22:14:16Z
C:\Tools\RegRipper4.0-main\rr.exe (3)
2026-04-05 22:06:39Z
Microsoft.Windows.Explorer (82)
2026-04-05 22:06:29Z
[S0089377-6AF8-4448-8957-A3773F02200E]\AccessData\FTK Imager\FTK Imager.exe (10)
2026-04-05 01:07:13Z
C:\Users\admin123\Downloads\WinTools\RegRipper4.0-main\rr.exe (1)
2026-04-05 01:05:00Z
Chrome (15)
2026-04-05 00:44:49Z
Microsoft.WindowsStore_Bueky3d8bbweApp (1)
2026-04-05 00:41:46Z
NSEdge (3)
2026-04-05 00:39:58Z
C:\Tools\Exterro_FTK_Imager_(x64).exe (2)
2026-04-04 22:21:00Z
Microsoft.XboxGamingOverlay_Bueky3d8bbweApp (19)
2026-03-14 14:42:40Z
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\voiceaccess.exe (1)
2026-03-14 14:30:12Z
[S0089377-6AF8-4448-8957-A3773F02200E]\WindowsApps\Microsoft.OutlookForWindows.1.0.0.0_neutral_Bueky3d8bbweApp (2)
2026-03-14 14:30:11Z
Microsoft.OutlookForWindows_Bueky3d8bbweApp\Microsoft.OutlookForWindows (2)
2026-03-14 14:30:01Z
Microsoft.Paint_Bueky3d8bbweApp (6)
[S0089377-6AF8-4448-8957-A3773F02200E]\WindowsApps\Microsoft.Paint.11.2511.291.0_x64_Bueky3d8bbweApp\PaintApp\mpaint.exe (1)
2026-03-14 00:56:08Z
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\cmd.exe (11)
2026-03-14 00:54:52Z
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\UserAccountControlSettings.exe (1)
2026-03-19 16:29:25Z
[7C5A40E7-A0F8-40FC-B744-C0F2E0B9FA8E]\BreakPoint Software\Hex Workshop v5\hworks32.exe (6)
2026-01-14 00:43:15Z
C:\Users\admin123\Downloads\python-3.14.2-amd64.exe (4)
2026-01-14 00:34:06Z
C:\Users\admin123\AppData\Local\Package Cache\0ef3959c-4d0b-4cd2-9d49-307eaf0a2b7f\python-3.14.2-amd64.exe (2)
2026-01-14 00:33:53Z
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\msiexec.exe (4)
2026-01-03 02:01:22Z
Microsoft.AutoGenerated (923D0477-5846-686B-A659-0FCCD73851A8) (2)
2026-01-03 00:28:09Z
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\SystemPropertiesAdvanced.exe (5)
2025-12-07 13:59:32Z
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\WindowsPowerShell\1.0\powershell.exe (3)
2025-12-07 13:48:20Z
C:\Python\python.exe (1)
2025-12-07 02:46:24Z
C:\Tools\Volatility\bin\vol.exe (1)
2025-12-07 02:26:28Z
C:\Users\admin123\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.PythonManager_3847v3x7pw1n\python.exe (1)
2025-12-06 23:48:52Z
PythonSoftwareFoundation.PythonManager_3847v3x7pw1n\Python.Exe (4)
2025-12-06 00:38:35Z
C:\Tools\ImagerPro_B.2.0.26\vx64\Exterro_FTK_Imager_(x64).exe (1)
2025-12-06 00:16:03Z
E:\vboxadditions\VBoxWindowsAdditions.exe (1)
2025-12-05 14:36:56Z
Microsoft.WindowsFeedbackHub_Bueky3d8bbweApp (14)
Microsoft.MicrosoftStickyNotes_Bueky3d8bbweApp (11)
Microsoft.WindowsCalculator_Bueky3d8bbweApp (8)
Value names with no time stamps:
USER_COUNTER
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\reg.exe
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\vbox\FirstLogonAnim.exe
[1AC14E77-02E7-4E5D-B744-2EB1AE5198B7]\vbox\rgnupdt.exe
Microsoft.Windows.Client.CBS_cw5n1h2txyewy\CortanaUI
Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\Ap
E:\vboxadditions\VBoxWindowsAdditions-amd64.exe
C:\Users\admin123\Downloads\ChromeSetup.exe
C:\Users\admin123\AppData\Local\Temp\Google7320_1560731787\bin\updater.exe
Microsoft.DesktopAppInstaller_Bueky3d8bbweApp
Microsoft.WindowsTerminal_Bueky3d8bbweApp
Microsoft.Windows.ShellExperienceHost_cw5n1h2txyewy\Ap
C:\Users\admin123\AppData\Local\Temp\6C087746-AF45-47E6-BF45-47157738D089\Exterro_FTK_Imager_(x64).exe
C:\Users\admin123\AppData\Local\Temp\100A05BC-D300-476B-8B4B-0FF46C70B45\Exterro_FTK_Imager_(x64).exe
Microsoft.Windows.WindowsInstaller
Windows.ImmersiveControlPanel_cw5n1h2txyewy\Microsoft.Windows.ImmersiveControlPanel
C:\Users\admin123\AppData\Local\Temp\Google40852_288918642\bin\updater.exe
C:\Users\admin123\AppData\Local\Temp\7869C889-F389-40E5-8D49-43EA1E6A6C2D\python-3.14.2-amd64.exe
C:\Users\admin123\AppData\Local\Temp\5036A652-0039-407F-9C3B-C7380599A18D\python-3.14.2-amd64.exe
C:\Users\admin123\AppData\Local\Temp\0D15B787-D4A8-4C2E-AE63-F140786D4062\python-3.14.2-amd64.exe
Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\FullTrustApp
C:\Users\admin123\AppData\Local\Temp\rustup-visualstudioSHOHO\vs_setup.exe
C:\Users\admin123\AppData\Local\Temp\1b9fa83f68e083deb42\vs_bootstrapper_d15\vs_setup_bootstrapper.exe
Microsoft.VisualStudio.Installer
VisualStudio.740e374c
Microsoft.Windows.Client.CBS_cw5n1h2txyewy\CortanaUIFullTrustApp
C:\Users\admin123\AppData\Local\Temp\70989A74-4B73-42DA-8C43-C135E74430D\python-3.14.2-amd64.exe
C:\Users\admin123\AppData\Local\Temp\4E0B07B8-4044-43CB-A68A-17AEC055AEE0\python-3.14.2-amd64.exe
Microsoft.SkyDrive.Desktop
Microsoft.Windows.ControlPanel
D:\VBoxWindowsAdditions-amd64.exe
C:\Users\admin123\AppData\Local\Temp\5C977914-6A08-4C98-A1E0-14F24C981300\python-3.14.2-amd64.exe
Microsoft.Windows.Apprep.ChxApp_cw5n1h2txyewy\Ap
[F3A1C85A-E3CC-4A2E-AF9D-585A70090442]
[F4E57C4B-2036-4570-AB48-4438CFE3309F]
2026-04-05 22:06:39Z
[9E3995A8-1F9C-4F13-8B27-48B2486C7174]\TaskBar\File Explorer.lnk (80)
2026-04-05 22:06:29Z
C:\Users\Public\Desktop\Exterro FTK Imager.lnk (4)
2026-04-05 01:05:00Z
C:\Users\Public\Desktop\Google Chrome.lnk (15)
2026-03-14 14:51:58Z
[0139D44E-6AF8-40F2-B698-3DAFCAE6FFB8]\AccessData\FTK Imager\FTK Imager.lnk (6)
2026-03-14 14:42:40Z
[A77F5077-2E2B-44C3-A6A2-ABA601054A51]\Accessibility\VoiceAccess.lnk (1)
2026-03-14 00:56:08Z
[A77F5077-2E2B-44C3-A6A2-ABA601054A51]\System Tools\Command Prompt.lnk (11)
2026-01-19 16:29:29Z
[0139D44E-6AF8-40F2-B698-3DAFCAE6FFB8]\System Tools\Task Manager.lnk (2)
2025-12-07 13:59:32Z
[A77F5077-2E2B-44C3-A6A2-ABA601054A51]\Hex Workshop v5\Hex Workshop.lnk (6)
2025-12-06 00:18:26Z
[0139D44E-6AF8-40F2-B698-3DAFCAE6FFB8]\Windows PowerShell\Windows PowerShell.lnk (3)
2025-12-05 14:42:38Z
[A77F5077-2E2B-44C3-A6A2-ABA601054A51]\File Explorer.lnk (2)

```

c. Identify the MAC addresses associated with devices or drives used by this user.

- 80:6F:6F:6E:69:63
- F0:8A:00:00:00:00
- 08:00:27:8A:B2:A6
- 06:1B:9C:30:5F:9E

```
mp2 v.20200921
(NTUSER.DAT) Gets user's MountPoints2 key contents
MITRE: T1021.002 (user activity)

MountPoints2
Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2
LastWrite Time 2026-01-19 16:29:08Z

Remote Drives:

Volumes:
2026-03-28 16:22:10Z
    {1d7be79d-d1fc-11f0-a1b4-806e6f6e6963}
2026-01-19 16:29:14Z
    {b9197bcd-f553-11f0-a1cc-0800278ab2a6}
2026-01-05 02:53:23Z
    {d22d8531-e8dc-11f0-a1c4-0800278ab2a6}
2026-01-05 02:43:16Z
    {00ddd1bb-0000-0000-007e-f08a00000000}
    {29852ce9-e8c4-11f0-a1c3-0800278ab2a6}
2025-12-06 00:20:48Z
    {6026d50c-b071-4f01-976f-061b9c305f9e}
2025-12-06 00:15:06Z
    {1d7be79e-d1fc-11f0-a1b4-806e6f6e6963}

Drives:
2025-12-05 14:37:51Z - CPC

Unique MAC Addresses:
80:6E:6F:6E:69:63
F0:8A:00:00:00:00
08:00:27:8A:B2:A6
06:1B:9C:30:5F:9E
```

d. List the recent documents accessed by this user.

- Registry Files
- systemdata.txt
- SYSTEM
- softdata.txt
- samdata.txt
- samdata.log
- SAM
- SAM.LOG1.FileSlack
- src
- main.rs
- secdata.txt
- SOFTWARE
- SECURITY
- SYSTEM
- Downloads

- Tools
- WinTools.zip
- ms-gamingoverlay:///
- kglcheck/
- symbols
- Memsamples
- Test2.mem
- Test2.zip
- test1.E01
- C:\
- New folder
- windows.zip
- vol3
- volatility3-develop.zip
- windows
- \_\_init\_\_.py
- volatility3-develop
- vol.py
- HexEditor
- Disk Partition
- SysinternalsToolssuite
- vboxadditions
- AUTORUN.INF
- autorun.sh

```

recentdocs v.20200924
(NTUSER.DAT) Gets contents of user's RecentDocs key

RecentDocs
**All values printed in MRUList\MRUListEx order.
Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
LastWrite Time: 2026-04-05 22:15:36Z
27 = Registry Files
38 = systemdata.txt
37 = SYSTEM
34 = softdata.txt
30 = samdata.txt
29 = samdata.log
36 = SAM
35 = SAM.LOG1.FileSlack
21 = scc
20 = main.rs
32 = secdata.txt
33 = SOFTWARE
31 = SECURITY
0 = SYSTEM
28 = Downloads
5 = Tools
26 = WinTools.zip
1 = ms-gamingoverlay:///
23 = kglcheck/
16 = symbols
12 = Memsamples
24 = Test2.mem
25 = Test2.zip
22 = test1.E01
6 = C:\
19 = New folder
18 = windows.zip
17 = vol3
15 = volatility3-develop.zip
14 = windows
13 = __init__.py
8 = volatility3-develop
7 = vol.py
11 = HexEditor
10 = Disk Partition
9 = SysinternalsToolssuite
3 = vboxadditions
4 = AUTORUN.INF
2 = autorun.sh

```

- e. A Jump List is a dynamic menu that contains recently accessed files, tasks, or applications (e.g., Word or Excel files from the Start menu). Identify and describe the JumpList contents for this user.
- o File Explorer.lnk  
Last accessed – 2026-04-05 @22:06:39  
Indicates that the user frequently uses File Explorer from the taskbar

- Exterro FTK Imager.lnk  
Last Accessed - 2026-04-05 @22:06:29  
User launched Exterro from the .exe in file explorer or from a desktop shortcut
- Google Chrome.lnk  
Last Accessed – 2026-05-05 @01:05:00  
User launched Google Chrome from the Desktop shortcut or .exe in file explorer
- FTK Imager.lnk  
Last Accessed – 2026-03-14 @14:51:58  
FTK Imager was launched again, showing repeated usage of forensic tools. This launch however, was from a shortcut other than the desktop or direct .exe file
- VoiceAccess.lnk  
Last accessed - 2026-03-14 14:42:39  
Accessibility tool opened from Start Menu
- Command Prompt.lnk  
Last Accessed - 2026-03-14 00:56:08  
Shows CMD was launched from Start Menu -> System Tools
- Hex Workshop.lnk  
Last Accessed - 2026-01-19 16:29:29  
Indicates use of the Hex Workshop forensic editor
- Task Manager.lnk  
Last Accessed - 2025-12-07 13:59:32  
Opened from Start Menu -> System Tools.
- Windows PowerShell.lnk  
Last Accessed - 2025-12-06 00:18:26  
Shows PowerShell was launched from a shortcut.
- Microsoft Edge.lnk  
Last Accessed: 2025 - 12-05 14:36:41  
Indicates Edge was opened from a desktop/taskbar shortcut.

```
{F4E57C4B-2036-45F0-A9AB-443BCFE33D9F}
2026-04-05 22:06:39Z
{9E3995AB-1F9C-4F13-B827-48B24B6C7174}\TaskBar\File Explorer.lnk (80)
2026-04-05 22:06:29Z
C:\Users\Public\Desktop\Exterro FTK Imager.lnk (4)
2026-04-05 01:05:00Z
C:\Users\Public\Desktop\Google Chrome.lnk (15)
2026-03-14 14:51:58Z
{0139D44E-6AFE-49F2-8690-3DAFCAE6FFB8}\AccessData\FTK Imager\FTK Imager.lnk (6)
2026-03-14 14:42:39Z
{A77F5D77-2E2B-44C3-A6A2-ABA601054A51}\Accessibility\VoiceAccess.lnk (1)
2026-03-14 00:56:08Z
{A77F5D77-2E2B-44C3-A6A2-ABA601054A51}\System Tools\Command Prompt.lnk (11)
2026-01-19 16:29:29Z
{A77F5D77-2E2B-44C3-A6A2-ABA601054A51}\Hex Workshop v5\Hex Workshop.lnk (6)
2026-01-03 02:01:22Z
{0139D44E-6AFE-49F2-8690-3DAFCAE6FFB8}\System Tools\Task Manager.lnk (2)
2025-12-07 13:59:32Z
{A77F5D77-2E2B-44C3-A6A2-ABA601054A51}\Windows PowerShell\Windows PowerShell.lnk (3)
2025-12-06 00:18:26Z
C:\Users\Public\Desktop\Microsoft Edge.lnk (1)
2025-12-05 14:42:38Z
{A77F5D77-2E2B-44C3-A6A2-ABA601054A51}\File Explorer.lnk (2)
```



### Q3. 15 Points

Using the *WinPrefetchView* (*WinPrefetch utility*), analyze the provided Prefetch folder.

Note: By default, the tool loads your system's Prefetch files. To analyze the provided prefetch folder, go to:

Options → Advanced Options and select the given Prefetch folder.

After loading the dataset:

1. Locate all Prefetch entries related to the *curl utility*.
2. Examine each entry carefully.

Answer the following:

- How many times was the curl utility executed in total? If multiple Prefetch files exist for curl, include all of them in your count.
  - CURL.EXE-392959E8.pf | Run Count: 1 | Path: C:\Windows\System32\curl.exe
  - CURL.EXE-DD3881F2.pf | Run Count: 3 | Path: C:\CURL\BIN\curl.exe
- Which specific Prefetch file(s) did you use to determine this?
  - The prefetch files used to determine this were CURL.EXE-392959E8.pf and CURL.EXE-DD3881F2.pf as they are the only two CURL related files in the dataset.

|                                                                                                        |                                          |       |          |                              |   |                                                               |     |
|--------------------------------------------------------------------------------------------------------|------------------------------------------|-------|----------|------------------------------|---|---------------------------------------------------------------|-----|
|  CURL.EXE-392959E8.pf | 1/2/2026 6:30:36 ... 4/26/2026 8:32:3... | 1,271 | CURL.EXE | C:\Windows\System32\curl.exe | 1 | 1/2/2026 7:30:37 PM                                           | No  |
|  CURL.EXE-DD3881F2.pf | 4/5/2026 8:44:00 ... 4/26/2026 8:32:3... | 5,646 | CURL.EXE | C:\CURL\BIN\CURL.EXE         | 3 | 4/5/2026 8:44:00 PM, 4/5/2026 8:43:58 PM, 4/5/2026 8:43:54 PM | Yes |

### Q4. 25 Points

Jimmy is suspicious that his computer might be hacked. He performed a memory dump of his system and ran several Volatility commands. The outputs below were generated from different Volatility plugins. Study the outputs carefully and determine what is wrong with Jimmy's computer. There may be rogue processes running, suspicious parent-child relationships, unexpected network connections, abnormal numbers of process instances, and misleading process names. Correlate the evidence across the plugin outputs. Find at least 6 such instances.

#### psscan Output

Volatility Foundation Volatility Framework 2.6

| Offset (P)         | Name        | PID | PPID | PDB       | Time created                 |
|--------------------|-------------|-----|------|-----------|------------------------------|
| Time exited        |             |     |      |           |                              |
| 0x000000003e2d7060 | System      | 4   | 0    | 0x1a3c000 | 2026-04-02 13:41:18 UTC+0000 |
| 0x000000003f09d4e0 | smss.exe    | 340 | 4    | 0x1b22000 | 2026-04-02 13:41:19 UTC+0000 |
| 0x000000003f1a01c0 | smss.exe    | 404 | 4    | 0x1b2d000 | 2026-04-02 13:41:20 UTC+0000 |
| 0x000000003f22b120 | smss.exe    | 468 | 4    | 0x1b39000 | 2026-04-02 13:41:21 UTC+0000 |
| 0x000000003f2d3b40 | csrss.exe   | 548 | 404  | 0x1c10000 | 2026-04-02 13:41:22 UTC+0000 |
| 0x000000003f35f060 | wininit.exe | 620 | 404  | 0x1c85000 | 2026-04-02 13:41:24 UTC+0000 |

|                    |                |      |      |           |            |          |          |
|--------------------|----------------|------|------|-----------|------------|----------|----------|
| 0x000000003f3fela0 | csrss.exe      | 628  | 468  | 0x1cad000 | 2026-04-02 | 13:41:25 | UTC+0000 |
| 0x000000003f4870c0 | winlogon.exe   | 700  | 468  | 0x1d02000 | 2026-04-02 | 13:41:27 | UTC+0000 |
| 0x000000003f52a1e0 | services.exe   | 780  | 620  | 0x1d81000 | 2026-04-02 | 13:41:29 | UTC+0000 |
| 0x000000003f5aef80 | lsass.exe      | 804  | 620  | 0x1dd0000 | 2026-04-02 | 13:41:30 | UTC+0000 |
| 0x000000003f640a20 | svchost.exe    | 940  | 780  | 0x1e41000 | 2026-04-02 | 13:41:41 | UTC+0000 |
| 0x000000003f6bba20 | svchost.exe    | 988  | 780  | 0x1eb2000 | 2026-04-02 | 13:41:43 | UTC+0000 |
| 0x000000003f74a460 | explorer.exe   | 1504 | 1468 | 0x20e1000 | 2026-04-02 | 13:43:15 | UTC+0000 |
| 0x000000003f7b8a20 | explorer.exe   | 1612 | 1576 | 0x2172000 | 2026-04-02 | 13:44:02 | UTC+0000 |
| 0x000000003f843860 | explorer.exe   | 1720 | 1688 | 0x2203000 | 2026-04-02 | 13:44:27 | UTC+0000 |
| 0x000000003f8c61e0 | notepad.exe    | 1884 | 1612 | 0x2280000 | 2026-04-02 | 14:05:10 | UTC+0000 |
| 0x000000003f940720 | cmd.exe        | 1948 | 1612 | 0x22f2000 | 2026-04-02 | 14:06:42 | UTC+0000 |
| 0x000000003f9b0f20 | chrome.exe     | 2056 | 1612 | 0x2366000 | 2026-04-02 | 14:07:19 | UTC+0000 |
| 0x000000003fa41980 | scvhost.exe    | 4120 | 1884 | 0x2415000 | 2026-04-02 | 14:08:01 | UTC+0000 |
| 0x000000003fac0240 | svch0st.exe    | 4216 | 780  | 0x2480000 | 2026-04-02 | 14:08:12 | UTC+0000 |
| 0x000000003fb4c660 | conhost.exe    | 4308 | 1948 | 0x24f7000 | 2026-04-02 | 14:08:15 | UTC+0000 |
| 0x000000003fbdff40 | System         | 888  | 780  | 0x2579000 | 2026-04-02 | 14:08:30 | UTC+0000 |
| 0x000000003fc5a220 | rundll32.exe   | 4472 | 4120 | 0x25f1000 | 2026-04-02 | 14:09:03 | UTC+0000 |
| 0x000000003fce4040 | powershell.exe | 4556 | 1948 | 0x2669000 | 2026-04-02 | 14:10:21 | UTC+0000 |

## pstree Output

Volatility Foundation Volatility Framework 2.6

| Name                                | Pid  | PPid | Thds | Hnds | Time       |
|-------------------------------------|------|------|------|------|------------|
| -----                               |      |      |      |      |            |
| -----                               |      |      |      |      |            |
| 0x000000003e2d7060:System           | 4    | 0    | 98   | 612  | 2026-04-02 |
| 13:41:18 UTC+0000                   |      |      |      |      |            |
| . 0x000000003f09d4e0:smss.exe       | 340  | 4    | 2    | 29   | 2026-04-02 |
| 13:41:19 UTC+0000                   |      |      |      |      |            |
| . 0x000000003f1a01c0:smss.exe       | 404  | 4    | 2    | 31   | 2026-04-02 |
| 13:41:20 UTC+0000                   |      |      |      |      |            |
| .. 0x000000003f2d3b40:csrss.exe     | 548  | 404  | 9    | 482  | 2026-04-02 |
| 13:41:22 UTC+0000                   |      |      |      |      |            |
| .. 0x000000003f35f060:wininit.exe   | 620  | 404  | 3    | 79   | 2026-04-02 |
| 13:41:24 UTC+0000                   |      |      |      |      |            |
| ... 0x000000003f52a1e0:services.exe | 780  | 620  | 7    | 214  | 2026-04-02 |
| 13:41:29 UTC+0000                   |      |      |      |      |            |
| .... 0x000000003f640a20:svchost.exe | 940  | 780  | 13   | 326  | 2026-04-02 |
| 13:41:41 UTC+0000                   |      |      |      |      |            |
| .... 0x000000003f6bba20:svchost.exe | 988  | 780  | 11   | 301  | 2026-04-02 |
| 13:41:43 UTC+0000                   |      |      |      |      |            |
| .... 0x000000003fac0240:svch0st.exe | 4216 | 780  | 4    | 62   | 2026-04-02 |
| 14:08:12 UTC+0000                   |      |      |      |      |            |
| .... 0x000000003fbdff40:System      | 888  | 780  | 2    | 41   | 2026-04-02 |
| 14:08:30 UTC+0000                   |      |      |      |      |            |
| ... 0x000000003f5aef80:lsass.exe    | 804  | 620  | 9    | 587  | 2026-04-02 |
| 13:41:30 UTC+0000                   |      |      |      |      |            |
| . 0x000000003f22b120:smss.exe       | 468  | 4    | 2    | 30   | 2026-04-02 |
| 13:41:21 UTC+0000                   |      |      |      |      |            |
| .. 0x000000003f3fela0:csrss.exe     | 628  | 468  | 10   | 501  | 2026-04-02 |
| 13:41:25 UTC+0000                   |      |      |      |      |            |
| .. 0x000000003f4870c0:winlogon.exe  | 700  | 468  | 4    | 117  | 2026-04-02 |
| 13:41:27 UTC+0000                   |      |      |      |      |            |
| ... 0x000000003f74a460:explorer.exe | 1504 | 1468 | 24   | 812  | 2026-04-02 |
| 13:43:15 UTC+0000                   |      |      |      |      |            |
| ... 0x000000003f7b8a20:explorer.exe | 1612 | 1576 | 27   | 896  | 2026-04-02 |
| 13:44:02 UTC+0000                   |      |      |      |      |            |
| .... 0x000000003f8c61e0:notepad.exe | 1884 | 1612 | 3    | 104  | 2026-04-02 |

```

14:05:10 UTC+0000
..... 0x000000003fa41980:scvhost.exe          4120    1884      6      131  2026-04-02
14:08:01 UTC+0000
..... 0x000000003fc5a220:rundll32.exe          4472    4120      5       94  2026-04-02
14:09:03 UTC+0000
.... 0x000000003f940720:cmd.exe                1948    1612      2       58  2026-04-02
14:06:42 UTC+0000
..... 0x000000003fb4c660:conhost.exe           4308    1948      2       47  2026-04-02
14:08:15 UTC+0000
..... 0x000000003fce4040:powershell.exe        4556    1948      7      123  2026-04-02
14:10:21 UTC+0000
.... 0x000000003f9b0f20:chrome.exe             2056    1612     19      731  2026-04-02
14:07:19 UTC+0000
... 0x000000003f843860:explorer.exe            1720    1688     25      841  2026-04-02
13:44:27 UTC+0000

```

## sessions Output

Volatility Foundation Volatility Framework 2.6

```

Session(V)      ID  Processes
-----
0xffffffffa80031b9060 0 4:System, 340:smss.exe, 404:smss.exe, 468:smss.exe,
                    548:csrss.exe, 620:wininit.exe, 780:services.exe, 804:lsass.exe,
                    888:System, 940:svchost.exe, 988:svchost.exe, 4216:svchost.exe
0xffffffffa80042c4060 1 628:csrss.exe, 700:winlogon.exe, 1504:explorer.exe
0xffffffffa80043d1060 2 1612:explorer.exe, 1884:notepad.exe, 1948:cmd.exe,
                    2056:chrome.exe, 4120:scvhost.exe, 4308:conhost.exe,
                    4472:rundll32.exe, 4556:powershell.exe
0xffffffffa80044eb060 3 1720:explorer.exe

```

## netscan Output

Volatility Foundation Volatility Framework 2.6

| Offset(P)          | Proto                        | Local Address   | Foreign Address      | State       |
|--------------------|------------------------------|-----------------|----------------------|-------------|
| Pid                | Owner                        | Created         |                      |             |
| 0x000000003d144010 | TCPv4                        | 0.0.0.0:135     | 0.0.0.0:0            | LISTENING   |
| 940 svchost.exe    | 2026-04-02 13:41:45 UTC+0000 |                 |                      |             |
| 0x000000003d144b80 | TCPv4                        | 0.0.0.0:445     | 0.0.0.0:0            | LISTENING   |
| 4 System           | 2026-04-02 13:41:20 UTC+0000 |                 |                      |             |
| 0x000000003d145120 | TCPv4                        | 10.0.2.15:49722 | 185.199.110.153:5555 | ESTABLISHED |
| 4120 scvhost.exe   | 2026-04-02 14:08:22 UTC+0000 |                 |                      |             |
| 0x000000003d1457a0 | TCPv4                        | 10.0.2.15:49723 | 91.219.236.10:5555   | SYN_SENT    |
| 4120 scvhost.exe   | 2026-04-02 14:08:29 UTC+0000 |                 |                      |             |
| 0x000000003d146020 | TCPv4                        | 10.0.2.15:49724 | 203.0.113.77:5555    | SYN_SENT    |
| 4120 scvhost.exe   | 2026-04-02 14:08:34 UTC+0000 |                 |                      |             |
| 0x000000003d146700 | TCPv4                        | 10.0.2.15:49740 | 172.217.5.110:443    | ESTABLISHED |
| 2056 chrome.exe    | 2026-04-02 14:07:27 UTC+0000 |                 |                      |             |
| 0x000000003d146ec0 | TCPv4                        | 10.0.2.15:49751 | 198.51.100.44:4444   | ESTABLISHED |
| 4216 svchost.exe   | 2026-04-02 14:08:44 UTC+0000 |                 |                      |             |
| 0x000000003d1476a0 | TCPv4                        | 127.0.0.1:5555  | 127.0.0.1:49801      | ESTABLISHED |
| 4472 rundll32.exe  | 2026-04-02 14:09:15 UTC+0000 |                 |                      |             |
| 0x000000003d147e80 | UDPv4                        | 0.0.0.0:68      | *:*                  |             |
| 988 svchost.exe    | 2026-04-02 13:41:44 UTC+0000 |                 |                      |             |

There are multiple red flags within Jimmy's system, many more than 6, some of which show that remote access has likely been established within his system, and that there is likely to be malware present on the system. The following red flags were established and have been separated by each scan type.

#### psscan

- Multiple smss.exe processes
  1. psscan shows 3 smss.exe processes running simultaneously
  2. There should only be one smss.exe process running as the Session Manager
  3. Due to the multiple instances, it is likely that there is malware impersonating smss.exe occurring
- Multiple explorer.exe processes
  1. psscan shows 3 explorer.exe processes
  2. A single logged on user likely only has one explorer.exe process running, however, since Jimmy is scanning his system using these commands, it is possible that he left multiple explorer windows open for results
  3. If Jimmy was not the only user with open explorer.exe processes however, this suggests that hidden sessions or malware persistence is occurring
- Fake process named scvhost.exe
  1. psscan lists a process named scvhost.exe
  2. This was a misspelling by an attacker who attempted to impersonate **svchost.exe**, a legitimate Windows process
- Fake process named svch0st.exe
  1. psscan lists a process named svch0st.exe
  2. This is another malware impersonation attempt by an attacker but was misspelled with a zero instead of "o".
- Second "System" process
  1. psscan shows two System processes with different PID's
  2. Windows should only have one System process running, this suggests that malware is again attempting to disguise itself

#### pstree

- Impossible parent-child relationship: notepad.exe has child scvhost which then has child rundll32.exe
  1. svchost cannot be run by Notepad as it's a service-host process
  2. scvhost.exe is another misspelled version of svchost
  3. The fake svchost.exe is launching rundll32.exe which looks like a malware execution chain

#### sessions

- Fake svch0st.exe running in Session 0
  1. Session 0 is used only for core system processes

2. This indicates that privilege escalation has likely been successfully done by an attacker

nmap

- svchost.exe is making 3 outbound connections to port 5555 via the following connections
  1. 185.199.110.153:5555
  2. 91.219.236.10:5555
  3. 203.0.113.77:5555
  4. Port 5555 is commonly used for botnets and RATs
- svchost.exe is connected to 198.51.100.44:4444
  1. Established connection with port 4444
  2. Port 4444 is commonly used with malware such as Metasploit
- rundll32.exe is communicating with localhost:5555
  1. rundll32 should not connect to open networks
  2. rundll32 in this case is the child of the fake svchost.exe that was launched by notepad, this is likely communicating back to the fake svchost.exe which is then transmitting the data